Week 7-1

Calculus + Random Variable Problems

September 17, 2024

2110205 STATS COMP ENG

CHULA
COMPUTER
ΣNGINEERING

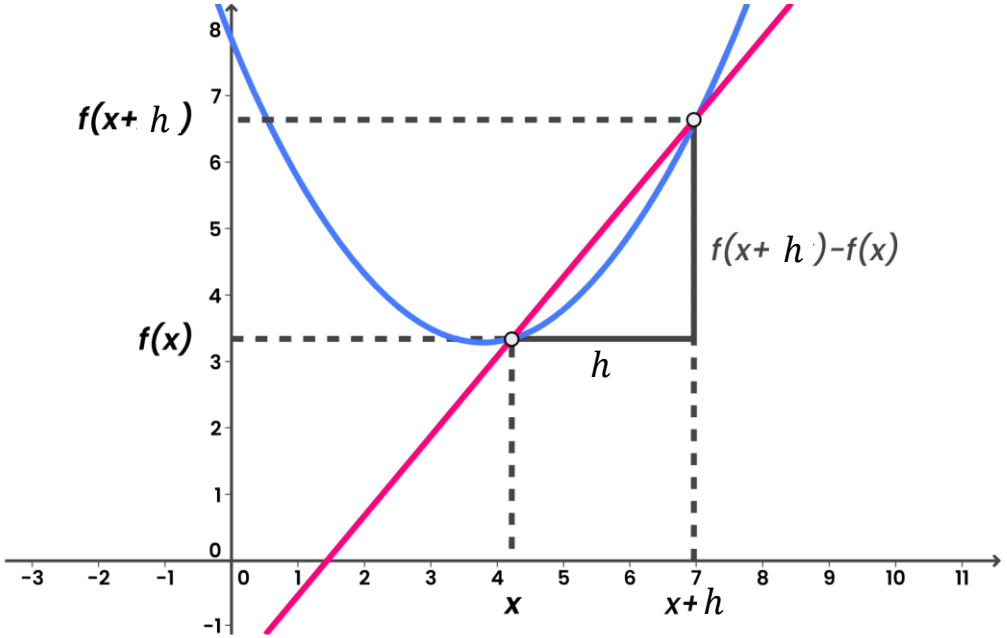
สรุปแคลคูลัสอีกรอบ

เฉพาะที่ใช้ในเนื้อหาวันนี้

กรุณาไปทบทวนกับสไลด์อาทิตย์ก่อนสำหรับเนื้อหาอื่นๆ

อนุพันธ์ของฟังชัน

- ความชันของกราฟ
- อัตราการเปลี่ยนแปลง



- อนุพันธ์ของฟังชัน $f(x)$ สามารถเขียนได้ว่า $f'(x)$ หรือ $\frac{d}{dx}f(x)$
- **นิยาม:**

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

สูตรอนุพันธ์พื้นฐาน

ควรทำให้ได้

$$\frac{d}{dx}(c) = 0 \qquad \frac{d}{dx}(x^n) = nx^{n-1}$$

$$(cf)' = cf' \qquad (f+g)' = f' + g' \qquad (f-g)' = f' - g'$$

$$(fg)' = fg' + gf' \qquad \left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$$

ตรีโกณ วิชานี้ไม่ใช้
แต่ COMP ENG MATH 2 ใช้

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

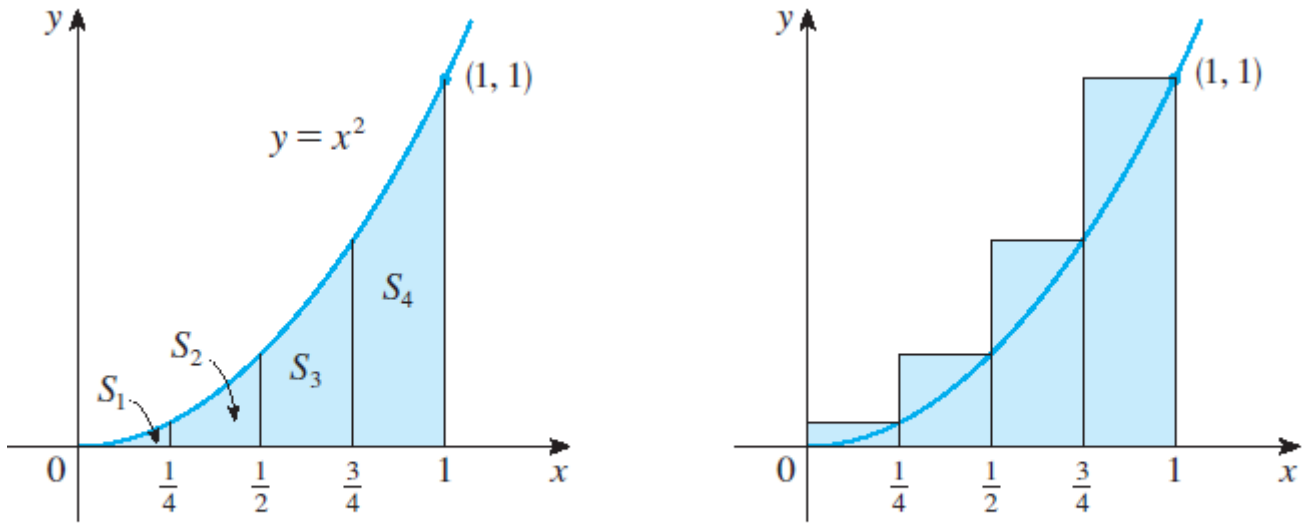
$$\frac{d}{dx}(\cos x) = -\sin x \qquad \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x \qquad \frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}e^x = e^x \qquad \frac{d}{dx}\ln x = \frac{1}{x}$$

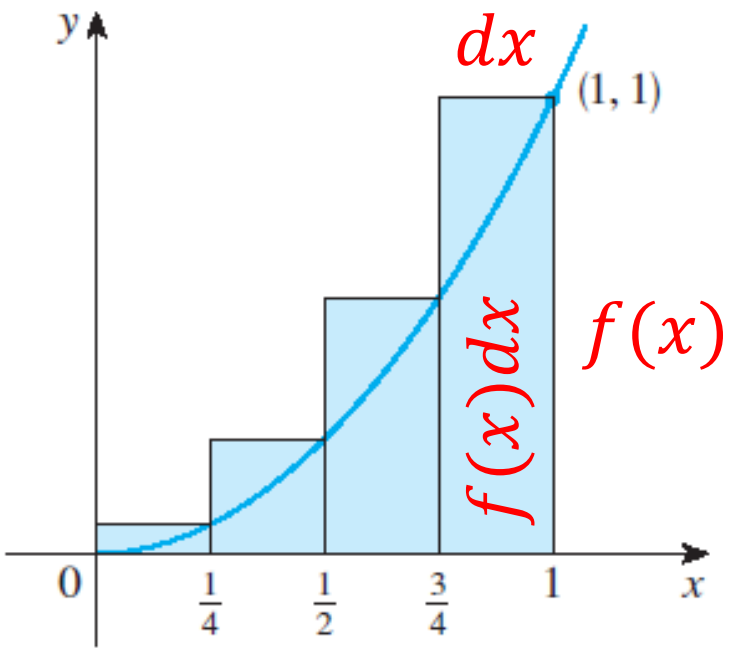
ปริพันธ์ของฟังชัน ในฐานะพื้นที่ใต้กราฟ

- **ปริพันธ์ (integral)** พื้นฐานเลยคือการคำนวณพื้นที่ใต้กราฟ
- เช่น กราฟ $y = x^2$ ต้องการหาพื้นที่ใต้กราฟตั้งแต่ $x = [0,1]$



ปริพันธ์ของฟังชัน ในฐานะการคูณกัน

- อีกมุมนึง มองได้ว่าคือการคูณกันของ $f(x)dx$.
- **dx แทน "ส่วนสั้นๆ ของ x"**
- ไอเดียคล้ายๆ กับ $\sum f(x)\,\Delta x$



สูตรปริพันธ์พื้นฐาน

ควรทำให้ได้

1 Table of Indefinite Integrals

$$\int cf(x)\,dx = c\int f(x)\,dx$$

$$\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

$$\int k\,dx = kx + C$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

$$\int \sec^2 x\,dx = \tan x + C$$

$$\int \csc^2 x\,dx = -\cot x + C$$

ตรีโกณ วิชานี้ไม่ใช้
แต่ COMP ENG MATH 2 ใช้

$$\int \sec x \tan x\,dx = \sec x + C$$

$$\int \csc x \cot x\,dx = -\csc x + C$$

$$\int e^x dx = e^x + C$$

$$\int \frac{1}{x} dx = \ln x + C$$

ปริพันธ์จำกัดเขต

- เวลาคำนวณ $\int f(x)dx$ เรียกว่า **ปริพันธ์ไม่จำกัดเขต** (Indefinite integral)
- ถ้าเป็น $\int_a^b f(x)dx$ เรียกว่า **ปริพันธ์จำกัดเขต** (Definite integral)
  - **ความหมาย:** หาพื้นที่ได้กราฟ $f(x)$ ตั้งแต่ $x = a$ ถึง b.
  - **หรือ**: $f(x)$ คูณ dx ตั้งแต่ $x = a$ ถึง b.

- **ความสัมพันธ์**: ถ้า $F(x) = \int f(x)dx$ แล้ว $\int_a^b f(x)dx = F(b) - F(a)$.

ปริพันธ์แยกส่วน

ควรทำให้ได้
แบบกำหนด u, v

- $\int u dv = uv - \int v du$
- ถ้าเราสามารถหา u และ v ที่เข้ารูปได้ เราสามารถใช้สูตรปริพันธ์แยกส่วนเพื่อคำนวณแทนได้

ตัวอย่าง: $\int x \cos x \, dx$ เมื่อ $u = x, v = \sin x$

ได้ $du = dx, dv = \cos x \, dx$

เพราะ $\int u dv = \int x \cos x \, dx$ ได้ตามโจทย์

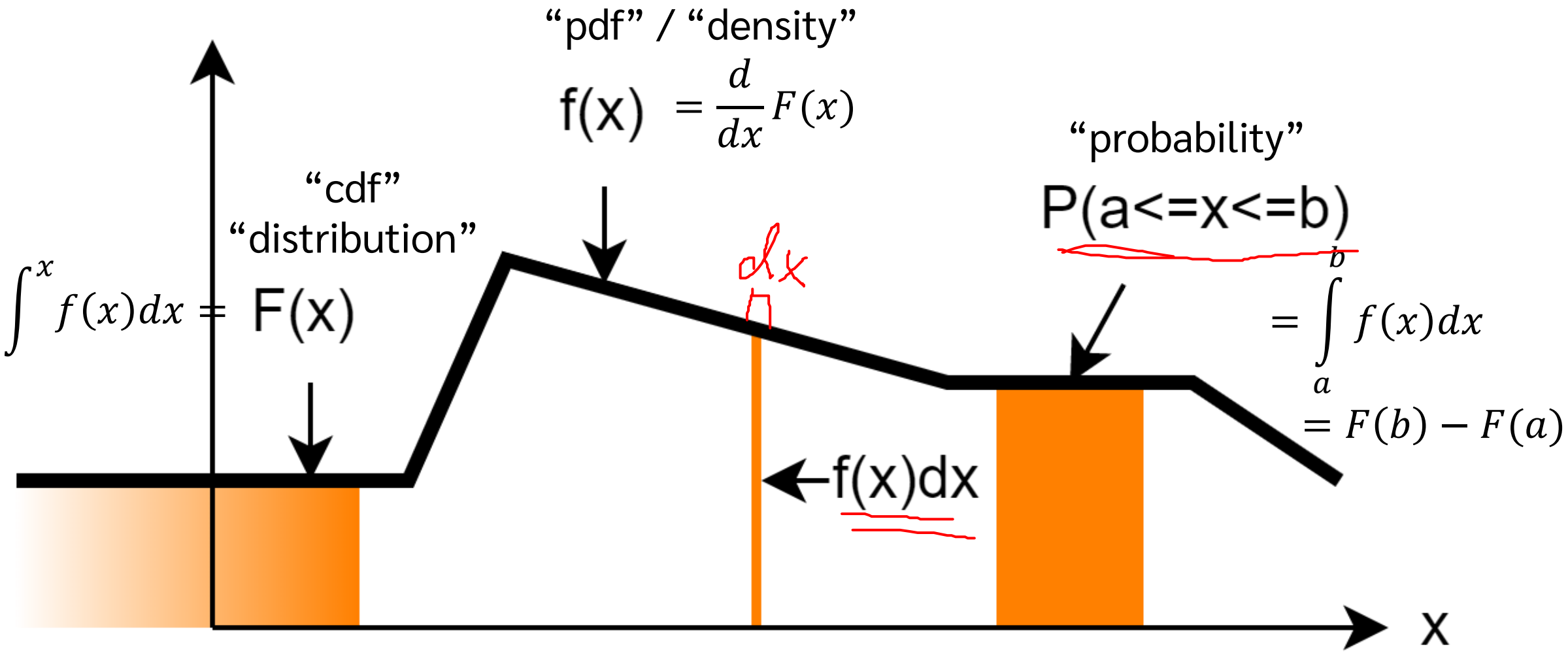
เลยได้ว่า $\int x \cos x \, dx = (x)(\sin x) - \int \sin x \, dx = x \sin x + \cos x + C$

เช็คชื่อ

โจทย์: จงคำนวณ $\int_0^{10}(10-x)dx$

Manipulating Continuous Random Variable

RECAP: Continuous random variable



Change of Variable (again)

ควรทำให้ได้

- Remember in Discrete Random Variable:
  $Y = X^2$ when X represent a die face:

X	1	2	3	4	5	6
Y	1	4	9	16	25	36
p(a)	1/6	1/6	1/6	1/6	1/6	1/6

Value change but probability is the same!

- So, we can do $\boxed{f_X(x)dx} = \cdots = \boxed{f_Y(y)dy}$ by changing the formula from x to y.

$f_X(x)$ is pdf, but
$f_X(x)dx$ is probability!

Transformation of Random Variables

ควรทำให้ได้

Problem. Given random variable X with distribution $U(0,2)$. Find range, cdf, and pdf of $Y = X^2$. ($F_X(x) = \frac{x}{2}$, $f_X(x) = \frac{1}{2}$ in range $[0,2]$)

Answer:

$$y = x^2 \rightarrow dy = 2xdx \rightarrow \frac{dy}{2x} = dx \rightarrow \frac{dy}{2\sqrt{y}} = dx.$$

Then: $f_X(x)dx = \frac{1}{2}dx = \frac{dy}{4\sqrt{y}} = f_Y(y)dy.$

Therefore: $f_Y(y) = \frac{1}{4\sqrt{y}}$

Normal Change of Variable

ควรรู้ผลลัพธ์
วิธีทำไม่เป็นไร

Problem. Assume $X \sim N(\mu, \sigma^2)$. Show that $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$.

$$z = \frac{x-\mu}{\sigma}$$

$$dz = d\left(\frac{x-\mu}{\sigma}\right)$$

$$dz = \frac{\cancel{\sigma}\overbrace{\boxed{d(x-\mu)}}^{=dx-d\mu=dx-0} - (z-\mu)\overbrace{\boxed{d\sigma}}^{=0}}{\sigma^{\cancel{2}}}$$

$$dz = \frac{1}{\sigma}dx$$

$$dx = \sigma dz$$

$$f_X(x)dx = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}}e^{-\left(\frac{1}{2}\right)\left[\underbrace{\boxed{\frac{x-\mu}{\sigma}}}_{=z}\right]^2}\underbrace{\boxed{dx}}_{=\sigma dz}$$

$$= \frac{1}{\cancel{\sigma}\sqrt{2\pi}}e^{-\frac{z^2}{2}}\cancel{\sigma}dz$$

$$= \frac{1}{\sqrt{2\pi}}e^{-\frac{z^2}{2}}dz$$

$$f_Z(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$$

CDF, Mean, and Variance of Distributions

Uniform Distribution – CDF

ควรทำให้ได้
แบบกำหนด a, b

Problem. Given the pdf $f(x) = \frac{1}{b-a}$ in range $[a, b]$. Find cdf $F(x)$.

$$F(x) = \int_a^x \frac{dx}{b-a} \qquad (a \le x \le b)$$

$$= \frac{1}{b-a}[x]\Big|_a^x$$

$$= \frac{x-a}{b-a}.$$

Uniform Distribution – Mean

ควรทำให้ได้
แบบกำหนด a, b

Problem. Let $X \sim Uniform(a, b)$. Find $E(X)$.

pdf

$$E(X) = \int_a^b (x)\boxed{\left(\frac{1}{b-a}\right)}dx = \left(\frac{1}{b-a}\right)\int_a^b x dx$$

$$= \left(\frac{1}{b-a}\right) \cdot \frac{x^2}{2}\Big|_a^b$$

$$= \left(\frac{1}{b-a}\right) \cdot \left(\frac{b^2-a^2}{2}\right)$$

ผลต่างกำลังสอง

$$= \frac{b^2-a^2}{2\cdot(b-a)} = \frac{\boxed{\cancel{(b-a)}(b+a)}}{2\cdot\cancel{(b-a)}}$$

$$= \frac{1}{2}(b+a)$$

Uniform Distribution – Variance

ควรทำให้ได้
แบบกำหนด a, b

Problem. Let $X \sim Uniform(a, b)$. Find $Var(X)$.

$$Var(X) = E(X^2) - E(X)^2$$

$$= \int_a^b x^2 \left(\frac{1}{b-a}\right) dx - \left[\frac{b+a}{2}\right]^2$$

$$= \frac{1}{b-a} \cdot \frac{x^3}{3}\Big|_a^b - \left[\frac{b+a}{2}\right]^2$$

ผลต่างกำลังสาม

$$= \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} - \frac{(b+a)^2}{4} = \frac{\boxed{(b-a)(b^2+ab+a^2)}}{3(b-a)} - \frac{b^2+2ab+a^2}{4}$$

$$= \frac{b^2+ab+a^2}{3} - \frac{b^2-2ab+a^2}{4} = \frac{4b^2+4ab+4a^2-3b^3-6ab-3a^2}{12}$$

$$= \frac{b^2-2ab+a^2}{12} = \frac{(b-a)^2}{12}$$

10 MIN BREAK

Exponential Distribution – CDF

ควรทำให้ได้

Problem. Given the pdf $f(x) = \lambda e^{-\lambda x}$ in range $[0, \infty]$. Find cdf $F(x)$.

$$F(X) = \int_0^x \lambda e^{-\lambda x} dx$$

$$= \int_0^{-\frac{u}{\lambda}} \frac{\lambda e^u}{-\lambda} du$$

$$\boxed{\begin{aligned} u &= -\lambda x \\ du &= -\lambda dx \end{aligned}}$$

$$= -e^u \Big|_0^{-\frac{u}{\lambda}} = -e^{-\lambda x} \Big|_0^x$$

$$= \left(-e^{-\lambda x}\right) - \left(-e^0\right)$$

$$= 1 - e^{-\lambda x}$$

Exponential Distribution – Mean

ควรรู้ผลลัพธ์
วิธีทำไม่เป็นไร

Problem. Let $X \sim \exp(\lambda)$. Find $E(X)$.

$$E(X) = \int_0^\infty \boxed{x}\boxed{\lambda e^{-\lambda x} dx}$$

$$= \boxed{x}\boxed{\left(-e^{-\lambda x}\right)}\Big|_0^\infty - \int_0^\infty \boxed{-e^{-\lambda x}}\boxed{dx}$$

$$\boxed{\int u dv = uv - \int v du}$$

$u = x, du = dx,$
$v = -e^{-\lambda x}, dv = \lambda e^{-\lambda x} dx$

$$\boxed{\lim_{x\to\infty}\left(-e^{-\lambda x}\right) = 0}$$

$$= 0 + \int_0^\infty \frac{e^u}{-\lambda} du$$

$$\boxed{\begin{aligned} u &= -\lambda x \\ du &= -\lambda dx \end{aligned}}$$

$$= 0 - \frac{e^{-\lambda x}}{\lambda}\Big|_0^\infty = 0 - \left[\frac{0}{\lambda} - \frac{1}{\lambda}\right]$$

$$= 1/\lambda$$

Exponential Distribution – Variance

ควรรู้ผลลัพธ์
วิธีทำไม่เป็นไร

Problem. Let $X \sim \exp(\lambda)$. Find $Var(X)$.

$$Var(X) = E(X^2) - E(X)^2$$

$$= \boxed{E(X^2)} - \left(\frac{1}{\lambda}\right)^2$$

$$= \frac{2}{\lambda^2} - \frac{1}{\lambda^2}$$

$$= \frac{1}{\lambda^2}$$

$$E(X^2) = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$$

$$= -x^2 e^{-\lambda x}\Big|_0^\infty + \int_0^\infty 2x e^{-\lambda x} dx$$

$$\boxed{\begin{aligned} u &= x^2 \\ v &= -e^{-\lambda x} \end{aligned}}$$

$$= -x^2 e^{-\lambda x}\Big|_0^\infty - \frac{2x e^{-\lambda x}}{\lambda}\Big|_0^\infty + \int_0^\infty \frac{e^{-\lambda x}}{\lambda} dx$$

$$\boxed{\begin{aligned} u &= 2x \\ v &= -\frac{e^{-\lambda x}}{\lambda} \end{aligned}}$$

$$= \left[-x^2 e^{-\lambda x} - \frac{2x e^{-\lambda x}}{\lambda} - \frac{2e^{-\lambda x}}{\lambda^2}\right]\Big|_0^\infty$$

$$= [0 - 0] - [0 - 0] - \left[0 - \frac{2}{\lambda^2}\right] = \frac{2}{\lambda^2}$$

Normal Distribution – CDF

ควรทำไม่ได้

- **Not feasible to calculate.**

Normal Distribution – Mean

ควรรู้ผลลัพธ์
วิธีทำไม่เป็นไร

Problem. Let $Z \sim N(0,1)$. Find $E(Z)$.

Find $E(Z) = \int_{-\infty}^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$.

Because of symmetry of the function, the mean is 0 if the integral converge.

$$\int_0^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} z e^{-\frac{z^2}{2}} dz = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-u} du$$

$$\boxed{\begin{aligned} u &= \frac{z^2}{2} \\ du &= z dz \end{aligned}}$$

$$= \frac{1}{\sqrt{2\pi}} \left[-e^{-z^2/2}\right]_0^{\infty} = \frac{1}{\sqrt{2\pi}} \left(0 - (-1)\right) = \frac{1}{\sqrt{2\pi}}.$$

Similarly, $\int_{-\infty}^{0} z \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = -\frac{1}{\sqrt{2\pi}}$.

Normal Distribution – Variance

ควรรู้ผลลัพธ์
วิธีทำไม่เป็นไร

Problem. Let $Z \sim N(0,1)$. Find $Var(Z)$.

As $E(X) = 0$:

$$Var(Z) = E(Z^2) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-\frac{z^2}{2}} dz$$

$u = z$

$v = -e^{-\frac{z^2}{2}}$

$$= \frac{1}{\sqrt{2\pi}} \left(-ze^{-\frac{z^2}{2}} \Big|_{-\infty}^{\infty} \right) + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz$$

$\lim_{x \to \infty} -ze^{-z^2/2} = 0$

$= \int f(z)dz$ the entire range of pdf of z

$$= \frac{1}{\sqrt{2\pi}} (0 - 0) + 1 = 1$$

Normal Distribution – Mean and Variance

ควรทำให้ได้

Problem. Show that $N(\mu, \sigma^2)$ has mean of μ and variance of σ^2.

Previously, $N(0,1)$ has $\mu = 0$ and $\sigma^2 = 1$. **Also,** $\boxed{Z = \frac{X-\mu}{\sigma}} \sim N(0,1)$.

Thus, $E(X) = E(\boxed{\sigma Z + \mu}) = \sigma E(Z) + \mu = \mu$.

Similarly, $Var(X) = Var(\sigma Z + \mu) = \sigma^2 Var(Z) = \sigma^2$.

Applied Problems

Quiz

1. Suppose the average time of a Rock-Paper-Scissors game follows an exponential distribution with mean of 2 minutes.
   1. Write the pdf of the distribution.
   2. If the winner get paid $t^2 + 3t$ when t is the game length, write the integral for the pdf of the payout.
2. The average ice cream scoop weighs 110g with standard deviation of 10 g.
   1. What is the probability that one scoop exceed 125g?
   2. What is the standard deviation if you buy 3 scoops?

Solution (1)

1. Suppose the average time of a Rock-Paper-Scissors game follows an exponential distribution with mean of 2 minutes.
   1. Write the pdf of the distribution.
   2. If the winner get paid $t^2 + 3t$ when t is the game length, write the integral for the mean of the payout.

Solution (2)

2. The average ice cream scoop weighs 110g with standard deviation of 10 g.
   1. What is the probability that one scoop exceed 125g?
   2. What is the standard deviation if you buy 3 scoops?

MORE PROBLEMS

ในเว็บ MIT ที่เป็น Text หลักของวิชานี้ มีแบบฝึกหัดและข้อสอบ (ของ MIT) ให้ทำได้นะครับ ไปทำกันได้ถ้าต้องการโจทย์เพิ่ม

Assignments

COURSE HOME
SYLLABUS
CALENDAR
INSTRUCTOR INSIGHTS
READINGS
CLASS SLIDES
ASSIGNMENTS
EXAMS
STUDIO RESOURCES
RELATED RESOURCES

Listed in the following table are problem sets and solutions. For each problem set, there is also an interactive problem set checker. Students in the class were able to work on the assigned problems in the PDF file, then use the problem set checker to input each answer into a box and find out if the answer was correct or incorrect. Students could use the problem set checker as many times as they wanted to, and they weren't graded on this in any way. Another form of interactive assessment used in the course, Interactive Reading Questions, can be viewed on the Readings page.

PROBLEMS	INTERACTIVE PROBLEM SET CHECKER	SOLUTIONS
Probability		
Problem Set 1 (PDF)	Problem Set 1 Checker	Solutions to Problem Set 1 (PDF)
Problem Set 2 (PDF) R Tutorial: Run For Loops R Tutorial: Run Length Encoding	Problem Set 2 Checker	Solutions to Problem Set 2 (PDF)
Problem Set 3 (PDF)	Problem Set 3 Checker	Solutions to Problem Set 3 (PDF)
Problem Set 4 (PDF)	Problem Set 4 Checker	Solutions to Problem Set 4 (PDF)
Statistics		
Problem Set 5 (PDF)	Problem Set 5 Checker	Solutions to Problem Set 5 (PDF)
Problem Set 6 (PDF) Problem Set 6 R Code (R)	Problem Set 6 Checker	Solutions to Problem Set 6 (PDF)
Problem Set 7 (PDF)	Problem Set 7 Checker	Solutions to Problem Set 7 (PDF)
Problem Set 8 (PDF)	Problem Set 8 Checker	Solutions to Problem Set 8 (PDF)
Problem Set 9 (PDF)	Problem Set 9 Checker	Solutions to Problem Set 9 (PDF)

Exams

COURSE HOME
SYLLABUS
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EXAMS
STUDIO RESOURCES
RELATED RESOURCES

Listed in the following table are practice exam questions and solutions, and the exam questions and solutions. Additional materials for exam preparation can be found under the class sessions dedicated to exam review. Students were encouraged to prepare a 4x6 inch notecard to use for reference during each exam.

EXAMS	QUESTIONS	SOLUTIONS
1	Exam 1 Practice Questions I (PDF)	Solutions to Exam 1 Practice Questions I (PDF)
	Exam 1 Practice Questions II (PDF)	Solutions to Exam 1 Practice Questions II (PDF)
	Exam 1 Practice Questions: Long List (PDF)	Solutions to Exam 1 Practice Questions: Long List (PDF)
	Exam 1 (PDF)	Solutions to Exam 1 (PDF)
2	Exam 2 Practice Questions (PDF)	Solutions to Exam 2 Practice Questions (PDF)
	Exam 2 (PDF)	Solutions to Exam 2 (PDF)
Final	Final Exam Practice Questions (PDF) *These practice questions cover only the material taught in class sessions after Exam 2.*	Solutions to Final Exam Practice Questions (PDF)
	Final Exam (PDF)	Solutions to Final Exam (PDF)

https://dspace.mit.edu/bitstream/handle/1721.1/153490/18-05-spring-2014/contents/readings/index.htm

END OF CLASS